Chapter 13: Brain Metabolism and Blood Supply

Comprehensive Neuroscience Reference

Abstract

The brain is the body's most metabolically demanding organ, consuming 20% of total body energy despite comprising only 2% of body mass. This chapter examines how the brain meets its extraordinary energy demands through continuous glucose delivery and oxidative metabolism. We review glucose transport (GLUT1 at the BBB, GLUT3 in neurons), glycolytic pathways in astrocytes (producing 2 ATP net per glucose plus lactate), and oxidative phosphorylation in neurons (yielding 30–32 ATP per glucose). The astrocyte-neuron lactate shuttle (ANLS) hypothesis proposes activity-dependent coupling: active neurons stimulate astrocytic glycolysis, producing lactate that is exported to nearby neurons for oxidative metabolism. Alternative fuels (ketone bodies during fasting/ketogenic diet, producing 29 ATP per molecule; circulating lactate during exercise) extend the brain's metabolic repertoire. Cerebral blood flow is tightly regulated through multiple mechanisms: metabolic regulation (CO_2 sensitivity: 4% CBF change per mmHg $PaCO_2$), myogenic autoregulation (constant flow from 60–150 mmHg MAP), and neurovascular coupling (activity-dependent vasodilation via nitric oxide, prostaglandins, and pericyte dilation). The BOLD signal in fMRI reflects local hemodynamic changes coupled to neural activity. Acute ischemic stroke disrupts this delicate system, causing ATP depletion, excitotoxic cascade, and neuronal death in the core and salvageable penumbra. Brain temperature regulation and therapeutic hypothermia are examined as protective strategies after cardiac arrest or hypoxic-ischemic injury.

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1. Energy Demands of the Brain

The human brain is the most metabolically demanding organ:

- **Weight**: ∼2% of body mass (∼1.4 kg)
- **Energy consumption**: ∼20% of total body energy at rest (∼20 watts)
- **Oxygen consumption**: ∼20% of total O_2 (∼3.5 mL O_2/100g/min)
- **Glucose consumption**: ∼25% of total glucose (∼5.6 mg glucose/100g/min)
- **Blood flow**: ∼15% of cardiac output (∼750 mL/min; ∼50 mL/100g/min)

This disproportionate energy demand reflects the cost of maintaining ionic gradients (Na^+/K^+-ATPase accounts for ∼50% of brain ATP consumption), synaptic transmission (∼25%), and intracellular signaling, protein synthesis, and axonal transport (∼25%).

Even at "rest," the brain is highly active — the default mode network maintains metabolic rates nearly equivalent to those during focused cognitive tasks. The difference in metabolic demand between resting and maximal cognitive effort is only ∼5–10%.

2. Glucose Metabolism

2.1 Glucose as Primary Fuel

Glucose is the brain's primary energy substrate under normal conditions. The brain cannot store significant glycogen (astrocytes store small amounts — ∼5 μmol/g) and therefore depends on continuous blood glucose delivery.

Glucose transport: Blood glucose crosses the BBB via **GLUT1** transporters (on endothelial cells and astrocytes). Neurons primarily use **GLUT3** (higher affinity than GLUT1, ensuring neuronal glucose uptake even at low concentrations).

2.2 Metabolic Pathways

1. **Glycolysis** (cytoplasm): Glucose → pyruvate (2 ATP + 2 NADH per glucose). Occurs predominantly in astrocytes, which are more glycolytic than neurons.
2. **Oxidative phosphorylation** (mitochondria): Pyruvate → acetyl-CoA → TCA cycle → electron transport chain → ∼30–32 ATP per glucose. Neurons are primarily oxidative, depending on mitochondrial ATP production.
3. **Astrocyte-neuron lactate shuttle (ANLS)** (Pellerin & Magistretti, 1994): Glutamatergic synaptic activity → astrocytic glutamate uptake (via EAAT1/2) → stimulates astrocytic glycolysis → lactate production → lactate export via MCT1/4 → neuronal lactate import via MCT2 → conversion to pyruvate → oxidative metabolism. This model suggests activity-dependent metabolic coupling: active neurons receive extra fuel from surrounding astrocytes in an activity-dependent manner.

Note: The ANLS model is influential but debated. Some evidence suggests neurons can directly metabolize glucose and may prefer it over lactate under certain conditions. The reality

likely involves both direct neuronal glucose metabolism and lactate shuttling, with the relative contributions varying by brain region, metabolic state, and activity level.

2.3 Pentose Phosphate Pathway

A minor but critical pathway: glucose-6-phosphate → NADPH + ribose-5-phosphate. NADPH is essential for:

- **Glutathione regeneration** (antioxidant defense — neurons are especially vulnerable to oxidative stress)
- **Lipid synthesis** (for myelin and membrane maintenance)
- **Nitric oxide synthesis**

2.4 Glycogen

Astrocytes store limited glycogen (∼5 μmol/g vs. ∼500 μmol/g in liver), which serves as an emergency glucose reserve:

- Mobilized during intense neural activity and hypoglycemia
- Glycogen phosphorylase → glucose-1-phosphate → glycolysis → lactate → neurons
- Sleep promotes glycogen replenishment; prolonged wakefulness depletes astrocytic glycogen

3. Alternative Fuels

3.1 Ketone Bodies

During prolonged fasting, starvation, or ketogenic diet, the liver produces **ketone bodies** (β-hydroxybutyrate, acetoacetate, acetone) from fatty acid oxidation. Ketones cross the BBB via MCT1 transporters and can supply up to **60–70% of brain energy** during prolonged fasting (reducing glucose requirement from ∼120 g/day to ∼40 g/day).

Ketone metabolism: β-hydroxybutyrate → acetoacetate → acetoacetyl-CoA → acetyl-CoA → TCA cycle → ATP. Ketones are more efficient per oxygen molecule than glucose and produce less oxidative stress.

Neonatal brain: Heavily reliant on ketones from breast milk fat; ketogenic metabolism supports the high energy demands of rapid brain growth.

Therapeutic ketosis: The ketogenic diet is an established treatment for drug-resistant epilepsy (mechanism: multiple — enhanced inhibition via altered neurotransmitter balance, reduced neuronal excitability via KATP channel activation, anti-inflammatory effects, enhanced mitochondrial function). Investigated for Alzheimer's disease, Parkinson's disease, and brain tumors (glioblastoma — exploiting cancer cells' dependence on glucose/inability to metabolize ketones).

3.2 Lactate

Beyond the ANLS, circulating blood lactate (from exercising muscles or hepatic production) can cross the BBB and be used as neuronal fuel. Exercise-induced lactate may contribute to exer-

cise's beneficial effects on brain function and neuroplasticity (partially via BDNF upregulation triggered by lactate signaling).

4. Cerebral Blood Supply

4.1 Arterial Supply

The brain's blood supply comes from four arteries that form the **Circle of Willis** at the base of the brain:

Anterior circulation (internal carotid arteries — ∼70% of cerebral blood flow):

- **Anterior cerebral artery (ACA)**: Medial surfaces of frontal and parietal lobes; leg and foot motor/sensory areas. Occlusion → contralateral leg weakness, personality changes (medial frontal).
- **Middle cerebral artery (MCA)**: Lateral surfaces of frontal, parietal, and temporal lobes — the largest cerebral artery supplying the most clinically significant territory. Occlusion → contralateral face/arm weakness/numbness, aphasia (dominant hemisphere), neglect (non-dominant), homonymous hemianopia. The MCA is the most common site of ischemic stroke.
- **Anterior communicating artery**: Connects left and right ACAs. Most common site of Circle of Willis aneurysms.

Posterior circulation (vertebral arteries → basilar artery — ∼30% of CBF):

- **Posterior cerebral artery (PCA)**: Occipital lobe, inferior temporal lobe, thalamus, midbrain. Occlusion → contralateral homonymous hemianopia, memory impairment (if hippocampal branches affected).
- **Superior cerebellar artery (SCA)**: Superior cerebellum, pons
- **Anterior inferior cerebellar artery (AICA)**: Lateral pons, middle cerebellar peduncle
- **Posterior inferior cerebellar artery (PICA)**: Lateral medulla, inferior cerebellum. Occlusion → lateral medullary (Wallenberg) syndrome.
- **Posterior communicating arteries**: Connect anterior and posterior circulations.

The Circle of Willis provides collateral circulation — if one feeding artery is occluded gradually, blood can reroute through communicating arteries. However, the Circle is anatomically complete in only ∼25% of individuals; many people have hypoplastic or absent communicating segments.

4.2 Venous Drainage

Cerebral veins → dural venous sinuses (superior/inferior sagittal sinus, transverse sinus, sigmoid sinus) → internal jugular veins. The dural sinuses are formed within the dura mater and are valveless. **Arachnoid granulations** protrude into the superior sagittal sinus, absorbing CSF into the venous blood.

5. Cerebral Blood Flow Regulation

5.1 Autoregulation

The brain maintains relatively constant blood flow (∼50 mL/100g/min) across a wide range of systemic blood pressures (mean arterial pressure ∼60–150 mmHg) through **myogenic** (smooth muscle contraction in response to stretch) and **metabolic** (vasodilation in response to local metabolites) mechanisms. Beyond these limits:

- Below ∼60 mmHg MAP: Cerebral ischemia (symptoms begin)
- Above ∼150 mmHg: Breakthrough → cerebral edema, hypertensive encephalopathy

5.2 Chemical Regulation

- **CO_2**: The most potent cerebral vasodilator. Increased $PaCO_2$ → decreased pH → arteriolar smooth muscle relaxation → increased CBF (∼4% increase per mmHg rise in $PaCO_2$). Hyperventilation → decreased $PaCO_2$ → vasoconstriction → reduced CBF (used clinically to reduce intracranial pressure).
- **O_2**: Severe hypoxia (PaO_2 <50 mmHg) → vasodilation → increased CBF
- **Adenosine**: Released during metabolic activity → vasodilation via A2A receptors on vascular smooth muscle

5.3 Neurovascular Coupling (Functional Hyperemia)

Active brain regions receive increased blood flow within 1–5 seconds of neural activation. This is the physiological basis of **fMRI (BOLD signal)**.

Mechanism: A multi-cellular process involving neurons, astrocytes, and vascular smooth muscle/pericytes:

1. **Neural activity** → glutamate release → activates postsynaptic neurons and surrounding astrocytes
2. **Neuronal pathway**: Interneurons (especially NOS+ interneurons) release NO and VIP → arteriolar dilation
3. **Astrocytic pathway**: Glutamate → mGluR5 on astrocytes → Ca^{2+} increase → arachidonic acid release → prostaglandin E_2 and epoxyeicosatrienoic acids (EETs) → vasodilation of arterioles via astrocyte endfeet contacting blood vessels. Under some conditions, the pathway produces 20-HETE → vasoconstriction (metabolic state-dependent switch).
4. **Pericyte-mediated capillary dilation**: Pericytes (contractile cells wrapping capillaries) relax in response to local signals → capillary dilation may precede arteriolar dilation and account for the earliest component of the hemodynamic response.

BOLD (Blood-Oxygen-Level-Dependent) signal: Increased CBF during neural activity exceeds the increase in oxygen consumption → local decrease in deoxyhemoglobin (paramagnetic) relative to oxyhemoglobin (diamagnetic) → increased MRI signal. The BOLD signal is an indirect measure of neural activity, reflecting primarily synaptic input and local processing rather than action potential output.

6. Vulnerability to Metabolic Disruption

6.1 Ischemic Stroke

Interruption of blood flow → rapid ATP depletion → failure of Na^+/K^+-ATPase → depolarization → massive glutamate release → excitotoxicity → cell death.

Ischemic cascade:

1. **Core** (severe ischemia, <10 mL/100g/min): Neurons die within minutes (necrosis)
2. **Penumbra** (moderate ischemia, 10–25 mL/100g/min): Electrically silent but structurally intact neurons. Potentially salvageable with reperfusion (thrombolysis with tPA within 4.5 hours; mechanical thrombectomy within 24 hours for large vessel occlusion).
3. **Excitotoxic damage**: Excessive glutamate → NMDA receptor overactivation → Ca^{2+} overload → mitochondrial dysfunction, free radical production, protease/lipase/endonuclease activation → cell death (necrosis and apoptosis)
4. **Reperfusion injury**: Restoration of blood flow → reactive oxygen species burst, inflammatory cell infiltration → additional damage

6.2 Hypoglycemia

Blood glucose below ∼3 mmol/L → impaired brain function (confusion, seizures). Below ∼1 mmol/L → irreversible brain damage (particularly hippocampus, basal ganglia, cortex). Neurons in the hippocampus and cortex are most vulnerable; brainstem neurons are most resistant.

6.3 Carbon Monoxide Poisoning

CO binds hemoglobin with 250× greater affinity than O_2 → tissue hypoxia. Characteristic delayed neurological damage to **basal ganglia** (especially globus pallidus) and white matter, occurring days to weeks after exposure.

7. Brain Temperature Regulation

Brain temperature is ∼0.5–1°C higher than core body temperature due to intense metabolic activity. Cerebral blood flow acts as a cooling system. Brain temperature varies by:

- **Region**: Active areas are warmer (up to 1°C gradient)
- **State**: Increases during arousal and seizures; decreases during sleep and anesthesia
- **Clinical relevance**: Therapeutic hypothermia (32–34°C) is neuroprotective after cardiac arrest and neonatal hypoxic-ischemic encephalopathy — reduces metabolic demand, excitotoxicity, and inflammation. Fever worsens outcomes after stroke and TBI.

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